



Dose tailoring of tacrolimus based on a non-linear pharmacokinetic model in children with refractory nephrotic syndrome

Ling Li^{a,b,1,2}, Min Zhu^{b,c,1,2}, De-Yi Li^a, Hong-Li Guo^a, Ya-Hui Hu^a, Ze-Yue Xu^{a,b,2}, Xia Jing^a, Feng Chen^{a,*}, Fei Zhao^d, Yun-Man Li^b, Jing Xu^{a,*}, Zheng Jiao^{c,*}

^a Department of Pharmacy, Children's Hospital of Nanjing Medical University, Nanjing, China

^b School of Basic Medicine and Clinical Pharmacy, China Pharmaceutical University, Nanjing, China

^c Department of Pharmacy, Shanghai Chest Hospital, Shanghai, China

^d Department of Nephrology, Children's Hospital of Nanjing Medical University, Nanjing, China

ARTICLE INFO

Keywords:

Refractory nephrotic syndrome
Population pharmacokinetics
Michaelis–Menten model
Tacrolimus
CYP3A5

ABSTRACT

The population pharmacokinetics (PPK) of tacrolimus (TAC) in children with refractory nephrotic syndrome (RNS) have not been well-characterized. This study aimed to investigate the significant factors affecting the TAC PPK characteristics of children with RNS and to optimize the dosing regimen.

A total of 494 concentrations from 108 children were obtained from routine therapeutic drug monitoring between 2016 and 2018. Information regarding the demographic features, laboratory test results, genetic polymorphisms of CYP3A5 (*rs776746*) and co-therapy medications were collected. PPK analysis was performed using the nonlinear mixed-effects modelling (NONMEM) software and two modelling strategies (the linear one-compartment model and nonlinear Michaelis–Menten model) were evaluated and compared.

CYP3A5 genotype, weight, daily dose of TAC and daily dose of diltiazem were retained in the final linear model. The absorption rate constant (K_a) was set at 4.48 h^{-1} in the linear model, and the apparent clearance (CL/F) and volume of distribution (V/F) in the final linear model were 14.2 L/h and 172 L , respectively. CYP3A5 genotype, weight and daily dose of diltiazem were the significant factors retained in the final nonlinear model. The maximal dose rate (V_{max}) and the average steady-state concentration at half- V_{max} (K_m) in the final nonlinear model were 2.15 mg/day and 0.845 ng/ml , respectively. The nonlinear model described the pharmacokinetic data of TAC better than the linear model in children with RNS. A dosing regimen was proposed based on weight, CYP3A5 genotype and daily dose of diltiazem according to the final nonlinear PK model, which may facilitate individualized drug therapy with TAC.

1. Introduction

Nephrotic syndrome (NS) is one of the most common glomerular diseases in children, and it is characterized by hypoalbuminaemia ($<25\text{ g/L}$), mass of proteinuria (urine protein:creatinine ratio $\geq 2000\text{ mg/g}$, or $3+$ protein on urine dipstick), edema and hyperlipidaemia. The incidence of NS is 4.7 (range 1.15–16.9) per 100,000 children each year, depending on ethnicity and geographical location [1]. Oral glucocorticoids have been recognized as the first-line treatment for NS. Approximately 80% of children are sensitive to corticosteroid therapy, whereas 80%–90% of them undergo relapse. Among those who relapse, 50%

experience frequently relapsing NS (FRNS) or develop steroid-dependent NS (SDNS) [2]. Additionally, approximately 20% of children do not achieve complete remission and eventually develop steroid-resistant NS (SRNS) [3]. The treatment of refractory nephrotic syndrome (RNS), which includes FRNS, SDNS and SRNS, is challenging. Meanwhile, the prolonged steroid exposure in children will lead to obesity, growth inhibition, hypertension, diabetes, osteoporosis, cataract and other adverse reactions. In the treatment strategy, children with RNS often need to be combined with other immunosuppressants.

Tacrolimus (TAC), a steroid-sparing agent, is recommended for treatment of RNS with desirable efficacy and fewer adverse reactions

* Corresponding authors.

E-mail addresses: chenfeng18@njmu.edu.cn (F. Chen), njxujing@163.com (J. Xu), jiaozhen@online.sh.cn (Z. Jiao).

¹ These authors contributed equally to this work.

² Visiting graduate student from China Pharmaceutical University.

[4,5]. TAC has low oral bioavailability and it is highly bind to red blood cells, α_1 -acid glycoprotein and albumin in the blood [6]. The main metabolic enzymes of TAC are CYP3A isozymes in the liver and gut wall, predominantly CYP3A5 and CYP3A4 [7]. The immunosuppressive effects of TAC have been shown to selectively inhibit the activation and proliferation of CD4 T-helper lymphocytes, and TAC can also stabilize the actin cytoskeleton of podocytes and thereby reduce proteinuria [8–10].

Due to the narrow therapeutic window and high inter- and intra-individual variability in the pharmacokinetics (PK) of TAC, routine therapeutic drug monitoring (TDM) is used clinically to optimize its dosage [11–13]. However, traditional TDM provides no information about how to tailor the dosage of TAC under different clinical scenarios or about the optimal starting dose of TAC. Therefore, population PK (PPK) combined with Maximum a posteriori Bayesian (MAPB) estimation has become a powerful method for individualized dosage regimens [14].

Currently, to the best of our knowledge, there are five PPK studies of TAC in children with NS or RNS, as summarized in Table 1 [15–19]. It is worth noting that the studies by Wang D [16] and Huang et al. [19] found that TAC clearance was dose-dependent. Disease states of RNS in children may affect TAC metabolism. For example, gastrointestinal edema may affect drug absorption, and hypoalbuminaemia can reduce protein binding [20–22]. In addition, the absorption variability of TAC is large [23]. TAC is highly lipophilic, and it almost completely binds to erythrocytes in the blood. It also can be transported outside the intestine by P-glycoprotein. These might contribute to the nonlinear process of TAC. Therefore, we assumed that a nonlinear relationship might exist between the dose and changes in TAC steady-state pharmacokinetics. And we adopted a nonlinear Michaelis–Menten (MM) model in children with RNS to explore the nonlinearity of TAC PPK.

In addition, from the Table 1, the results of previous studies are not consistent. In children with NS, Wang XQ et al. established a TAC PPK model based on weight (WT) and age [17], Hao et al. found that the CYP3A5 genotype was a covariate other than WT [15], and Huang et al.

described a PPK model that included not only WT and CYP3A5 genotype but also combined use of azole antifungal agents and diltiazem [19]. In children with RNS, Wang D et al. established a TAC PPK model based on age, cystatin-C and daily dose of TAC [16], while the model established by Chen et al. was based on WT, CYP3A5 genotype and coadministration with Wuzhi capsule [18]. In addition, these studies had some limitations. For example, the number of patients was limited, and only one study investigated the effect of diltiazem on TAC metabolism. Moreover, three studies considered the genotype of CYP3A5, but they did not distinguish between the effects of wild-type (*1*1) and heterozygous (*1*3) CYP3A5 on TAC metabolism. The genotype of CYP3A5 is an important factor that affects TAC metabolism [24–26]. Diltiazem can inhibit CYP3A enzyme activity and thereby affect the metabolism of TAC [27,28]. At the same time, diltiazem can effectively prevent hypertension and improve renal function [29]. According to our previous report [30], in the treatment of children with RNS, diltiazem was well-tolerated, and for the majority of children in the study, the TAC concentration was adjusted by increasing or decreasing the dose of diltiazem.

Therefore, the present study was conducted to establish and compare the differences between linear and nonlinear PPK models of TAC in children with RNS; to evaluate the impact of the CYP3A5 genotype, daily dose of diltiazem, and patient characteristics on the TAC PK parameters; and to develop individualized dosing regimens based on the superior model.

2. Methods

2.1. Patients and data collection

This retrospective single-centre pharmacokinetic study was carried out in the Children's Hospital of Nanjing Medical University. A total of 108 children (72 boys and 36 girls) with RNS who were treated with TAC from January 2016 to December 2018 were enrolled (Fig. 1). All the enrolled children met the diagnostic criteria for NS (urine protein:

Table 1

Summary of Population pharmacokinetic studies on tacrolimus in children with nephrotic syndrome.

Number (boy/girl)	Age (year)	Weight (kg)	Disease	Detection method	Sampling	Pharmacokinetic parameters	BSV CL(%)	BSV V (%)	Refs.
28 (19/9)	9.5 ± 4.4	36.5 ± 17.4	NS	HPLC/MS	before and at 1, 2, 3, 6, 9 and 12 h after taking TAC	Ka = 5.21 CL/F = 30.9 × (WT/70) ^{0.75} × 1.6 (if CYP3A5*1*1 or *1*3) V/F = 411 × (WT/70)	43.8	99.4	[15]
65 (44/21)	7.61 ± 3.92	30.85 ± 17.12	RNS	EMIT	C ₀	Ka = 4.48 (fixed) CL/F = 5.46 × EXP(0.0323 × age) × EXP(-0.359 × CYSC) × EXP(0.148 × DD) V/F = 57.1	22.2	0.2	[16]
41 (32/9)	8.05 ± 3.68	30.53 ± 13.51	NS	EMIT	C ₀	Ka = 4.5 (fixed) CL/F = 17.7 × (WT/70) ^{0.75} × (age/8.3) ^{-0.26} V/F = 314 × (WT/70)	31.1	–	[17]
17 (15/2)	9.90 ± 3.24	34.76 ± 15.13	RNS	EMIT	C ₀	Ka = 4.48 (fixed) CL/F = 20.1 × (WT/70) ^{0.75} × 1.5 (if CYP3A5*1*1 or *1*3) × (1–0.303 × WZ) V/F = 1210 × (WT/70)	15.5	61.3	[18]
71 (51/20)	7.8 ± 3.7	28.4 ± 14.1	NS	EMIT	C ₀	Linear model: Ka = 4.48 (fixed) CL/F = 7.13 × (DD/1.5) ^{0.225} × (WT/25) ^{0.265} × 1.394 (if CYP3A5*1*3) × 0.538 (if co-therapy with azole antifungal agents) × 0.88 (if co-therapy with diltiazem) V/F = 142 Nonlinear Michaelis–Menten model: Vmax = 1.92 × (WT/25) ^{0.559} Km = 1.98 × (WT/25) ^{0.559} × 0.189 (if CYP3A5*1*3) × 4.12 (if co-therapy with azole antifungal agents) × 2.01 (if co-therapy with diltiazem)	23.9	–	[19]

Abbreviations: NS: nephrotic syndrome; RNS: refractory nephrotic syndrome, HPLC/MS: high-performance liquid chromatography with tandem mass spectrometry; EMIT: enzyme multiplied immunoassay technique; C₀: trough concentration of tacrolimus; WT: weight; CYSC: cystatin-C; DD: daily dose of tacrolimus; WZ: wuzhi-capsule; Ka: absorption rate constant; BSV: between-subject variability; CL/F: apparent oral clearance; V/F: apparent volume of distribution; Vmax: maximum daily dose at the steady-state concentration; Km: steady-state trough concentration at the half-maximum daily dose.

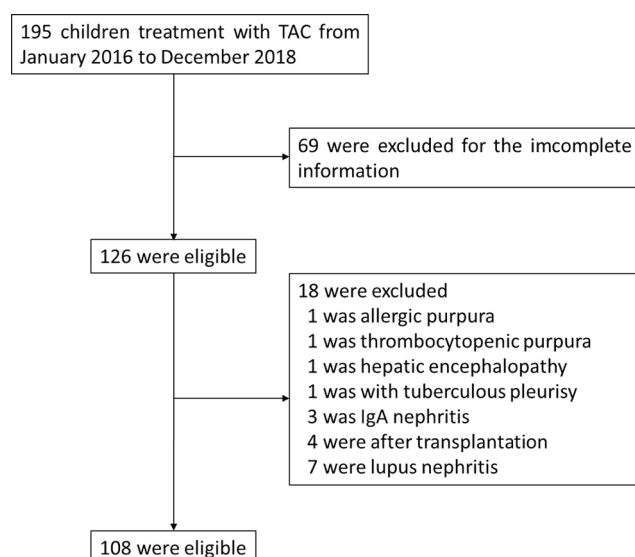


Fig. 1. Numbers of children who were eligible to be included in the study.

creatinine ratio ≥ 2000 mg/g or 3–4 + protein on urine dipstick, hypoalbuminaemia ≤ 25 g/L, and edema) [31,32]. RNS in this study included SRNS (failure to achieve complete remission after 4 weeks of corticosteroid therapy), SDNS (two consecutive relapses during corticosteroid therapy or within 14 days after discontinuation of therapy) and FRNS (two or more relapses within 6 months of the initial response, or four or more relapses within any 12 months) [31,32]. The exclusion criteria were: 1) children whose information was unable to be obtained and 2) children with serious infections or multiple organ injuries. The study was approved by the Ethics Committee of the Children's Hospital of Nanjing Medical University. Written consents were waived due to the retrospective nature of the study.

TAC was administered orally in the form of capsule (Prograf®, Astellas Ireland Co., Ltd., Kerry, Ireland). The initial dose of TAC was 0.05–0.15 mg/kg/day according to the guidelines set forth by the Nephrology Section of the Chinese Medical Association Pediatrics Branch in 2016. This dose was later adjusted based on TDM, clinical efficacy and adverse reactions. Each enrolled child received TAC treatment for at least 6 months. Diltiazem was usually given to increase the concentration of TAC, and the daily dose ranged from 1 to 4 mg/kg according to the data from routine TDM for TAC.

The demographic and biochemical characteristics of each child were extracted from the hospital internal system (HIS), including age, WT, sex, pathological classification, daily dose of TAC (DD), sampling time, daily dose of diltiazem (DILT), total bilirubin (TB), aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), gamma-glutamyl transpeptidase (GGT), albumin (ALB), serum creatinine (SCR), cystatin-C (CYSC), blood urea nitrogen (BUN) and uric acid (UA).

2.2. Analytical method of TAC concentration and genotyping

Whole blood samples (2 ml) for the determination of the trough concentration (C_0) were obtained before administration of the morning dose of TAC and were analysed using an Emit® 2000 Tacrolimus assay (SIEMENS, Munich, Germany). The experiment was conducted according to the manufacturer's instructions, and the calibration range was 2 to 30 ng/ml. The accuracy of the quality control samples was determined, and intra- and inter-assay imprecision were $<15\%$. At least 4 to 5 observations of the steady-state concentration were collected for each child.

Ethylenediaminetetraacetic acid anticoagulated blood was collected for genotyping of *CYP3A5 rs776746*. DNA was extracted by using a

TIANamp Blood DNA Kit (Tiagen Biotech, Beijing, China). The PCR primers were: 5'-TTGCCTCTTTGTACTTCTT-3' (forward) and 5'-GGTCTAGTTCATTAGGGTG-3' (reverse). The polymorphism of *CYP3A5 rs776746* was determined by direct automated DNA sequencing using an ABI 3730XL analyser after PCR (Applied Biosystems). A Chi-squared (χ^2) test was performed to assess the deviation of the genotype frequencies from Hardy–Weinberg equilibrium.

2.3. Population pharmacokinetic modelling of tacrolimus

The PPK of TAC was analyzed using the nonlinear mixed-effects model program (NONMEM®, Version 7.3; Icon Inc., PA, USA) compiled with gFortran (Version 4.9.2). R package version 3.6.1, and Xpose 4.5.3 were used to evaluate the model. The first-order conditional estimation with η - ϵ interaction (FOCE-I) was used throughout the model-building procedure to estimate the PK parameters and their variability.

2.3.1. Base model

For development of the model, we adopted a linear PK model and a nonlinear model (MM model) separately. In the linear PK model, since all of the TAC concentrations in this research were trough concentrations and observations of the absorption and distribution phases were lacking, a one-compartment model with first-order elimination was used to estimate the apparent clearance (CL/F) and apparent volume of distribution (V/F). The absorption rate constant (K_a) was set as 4.48 h^{-1} according to previously reports in the literature [16,33–35]. The sensitivity of this fixed parameter choice was also considered.

The MM model investigated is shown below:

$$\text{Dose} = \frac{V_{\max} \cdot C_{ss}}{K_m + C_{ss}} \quad (1)$$

Dose is the daily dose of TAC. $V_{\max}$ is the maximum daily dose at the steady-state concentration, and K_m is the steady-state trough concentration at the half-maximum daily dose. C_{ss} represents the steady-state concentration observed after at least 6 oral doses.

Interindividual variability was explored using an exponential error model as shown below:

$$P_i = T(P) \times \text{EXP}(\eta_i) \quad (2)$$

P_i is the individual parameter value, $T(P)$ is the typical individual parameter value, and η_i is the symmetrical distribution, which is a random term with a zero-mean and variance ω^2 .

The residual error variability was evaluated with additive, exponential and mixed error models. The equations were as follows:

$$Y = F + \epsilon \quad (3)$$

$$Y = F \times \text{EXP}(\epsilon) \quad (4)$$

$$Y = F \times \text{EXP}(\epsilon_1) + (\epsilon_2) \quad (5)$$

Y represents the observation, F is the individual prediction, and ϵ_n is the symmetrical distribution, which is a random term with a zero-mean and variance σ^2 .

2.3.2. Covariate models

The potential covariates included age, WT, sex, pathological classification, DD, TB, ALT, AST, ALP, GGT, ALB, SCR, CYSC, BUN, UA, *CYP3A5* polymorphisms (*rs776746*) and concomitant medications (diltiazem). Only the covariates present in $>10\%$ of all children in the study were investigated.

The categorical covariates (such as sex) were investigated with a proportional model (Eq. (6)). The continuous variables were investigated with a linear model (Eq. (7)), power function model (Eq. (8)) or exponential model (Eq. (9)). Considering that the dose of diltiazem was

different for each child, it was included both as a continuous variable (daily dose of diltiazem, DILT) and as a categorical variable (whether used when sampling).

$$P_i = \begin{cases} T(P) \times \theta & \text{if boy} \\ T(P) & \text{if girl} \end{cases} \quad (6)$$

$$P_i = T(P) + \theta \times \left(\frac{\text{COV}}{\text{COV}_m} \right) \quad (7)$$

$$P_i = T(P) \times \left(\frac{\text{COV}}{\text{COV}_m} \right)^\theta \quad (8)$$

$$P_i = T(P) \times \text{EXP}(\theta \times \left(\frac{\text{COV}}{\text{COV}_m} \right)) \quad (9)$$

where θ represents the degree of influence of the covariates on the parameters, COV is the covariate value for an individual and COV_m is the median value or typical population value of the covariate.

The covariate model was established in a stepwise way with forward inclusion and backward elimination. To compare the nested models, a likelihood ratio test was adopted. When a decrease in the objective function values (OFV) of more than 3.84 ($P < 0.05$, $df = 1$) was observed, the covariate was considered to be included in the model. After establishing a full covariates model, the model further backward eliminated covariates from each PK parameter one by one to obtain the final model. An increase in the OFV of at least 6.63 ($P < 0.01$, $df = 1$) was considered as the standard to retain significant covariates in the final model. The accuracy and rationality of parameter estimation were considered when selecting covariates. Physiological rationality was also considered.

2.3.3. Model evaluation

First, goodness-of-fit (GOF) plots was used for visual inspection, including: 1) observed value (DV) vs. population predicted value (PRED); 2) DV vs. individual predicted value (IPRED); 3) conditional weighted residuals (CWRES) vs. time after dose; and 4) CWRES vs. PRED.

Second, the stability and reliability of the final parameter estimates were evaluated by bootstrap, an internal validation method that was generated by repeated random sampling (1000 times) with replacement from the original database. This procedure was performed using the software Wings for NONMEM (Version 7.3; Icon Inc., PA, USA). The medians and 2.5% – 97.5% percentiles from the bootstrap consequences were compared with the final PK parameter estimates.

Third, the predictive performance of the final model was evaluated by normalized prediction distribution errors (NPDE). If the model is satisfied, the NPDE will follow a normal distribution (Shapiro-Wilks test) with a mean value of zero (t -test) and a variance of one (Fisher's test). Fourth, the final model was assessed using a predictive-correction visual predictive check (pcVPC), which was performed by the Perl-speaks-NONMEM program (PsN, Version 4.8).

Finally, the predictive performance of the linear and nonlinear models was compared by predictive-based diagnosis and Bayesian prediction. PRED and IPRED were estimated and compared with DV by estimating the relative prediction error (PE%, Eq. (10)) and individual prediction error (IPE%, Eq. (11)), respectively. IF₂₀ and IF₃₀, which indicate the percentage of IPE that fell within the range of $\pm 20\%$ and $\pm 30\%$, respectively, were used to represent combined indices of accuracy and precision.

$$\text{PE}(\%) = \frac{\text{PRED-DV}}{\text{DV}} \times 100 \quad (10)$$

$$\text{IPE}(\%) = \frac{\text{IPRED-DV}}{\text{DV}} \times 100 \quad (11)$$

2.4. Dosing regimen optimization

Monte Carlo simulations were carried out based on the parameter estimates from the final PPK model. The aim was to determine the optimal starting dosing regimen and achieve the target C₀ of 5–10 ng/ml. The simulation scenarios were designed according to the covariates included in the final model. Two thousand simulations were carried out in each clinical scenario using the initial dataset, and the steady-state C₀ of each simulated subject was calculated.

3. Results

3.1. Patient characteristics

A total of 108 children with RNS were enrolled, and 494 TAC trough concentrations were collected in this study. Each child was sampled at 1–3 TAC dose levels (1:2:3 = 26:75:7), and the TAC dose ranged from 0.02 to 0.13 mg/kg/day. The frequency of the CYP3A5 genotype in the population was in Hardy–Weinberg equilibrium. The numbers of children with the CYP3A5*1*1, *1*3 and *3*3 genotypes were 12, 50 and 46, respectively. About four-fifths of the children in the study received combined diltiazem treatment. The children's characteristics and co-medications are presented in Table 2.

3.2. Model building

3.2.1. Linear PK model

A one-compartment model with first-order elimination was chosen to describe the disposition of TAC. The residual error was best described by a mixed error model. The deviations in K_a didn't change the model fits, which further justify this model choice, and the detail results were shown in Table 3. The population typical values of CL/F and V/F in the

Table 2
Demographic and clinical characteristics of children (n = 108).

Characteristic	Mean $\pm$ SD	Median (range)
Sex (boys/girls)	108 (72/36)	
Number of TAC samplings	494	
Age (years)	6.26 $\pm$ 3.52	5.17 (1.25–15.83)
Weight (kg)	25.66 $\pm$ 12.98	20.5 (9–65)
TAC concentration (ng/ml)	3.97 $\pm$ 1.62	3.8 (2–10.6)
TAC daily dose (mg/day)	1.39 $\pm$ 0.51	1.5 (0.25–3.0)
TB (μ mol/L)	7.06 $\pm$ 3.32	6.5 (0.51–23.04)
AST (U/L)	20.18 $\pm$ 8.4	19 (7–77)
ALT (U/L)	14.90 $\pm$ 11.88	12 (3–120)
ALP (U/L)	187.52 $\pm$ 111.89	177 (45–1798)
GGT (50 U/L)	17.80 $\pm$ 11.27	15 (5–154)
ALB (g/L)	40.38 $\pm$ 8.76	43.15 (10.1–56.4)
SCR (μ mol/L)	32.68 $\pm$ 12.05	30.65 (13–126)
CYSC (mg/L)	1.03 $\pm$ 0.21	1.0 (0.53–1.87)
BUN (mmol/L)	4.20 $\pm$ 1.26	3.97 (0.5–9.6)
UA (μ mol/L)	303.61 $\pm$ 80.07	291.5 (131–667)
Pathological classification		
MCN	25 (23.15)	
MsPGN	49 (45.37)	
FSGS	22 (20.37)	
Missing	12 (11.11)	
Diltiazem		
Daily dose (mg/day)	34.14 $\pm$ 29.55	30 (0–90)
Whether used (yes/no)	87/21	
CYP3A5 rs776746		
AA (*1*1)	12 (11.1)	
AG (*1*3)	50 (46.3)	
GG (*3*3)	46 (42.6)	

Abbreviations: TAC, tacrolimus; TB, total bilirubin; AST, aspartate aminotransferase; ALT, alanine aminotransferase; ALP, alkaline phosphatase; GGT, gamma-glutamyl transpeptidase; ALB, albumin; SCR, serum creatinine; CYSC, Cystatin-C; BUN, blood urea nitrogen; UA, uric acid; MCN, minimal change nephropathy; MsPGN, mesangial proliferative glomerulonephritis; FSGS, focal segmental glomerulosclerosis; SD, standard deviation.

Table 3The effect of changing K_a in the linear model.

Ka	OFV	CL/F		V/F		$\omega_{CL/F}$ (%)		σ_1 (%)		σ_2	
		Estimate	RSE (%)	Estimate	RSE (%)	Estimate	RSE (%)	Estimate	RSE (%)	Estimate	RSE (%)
4.48	735.04	10.9	6.3	272	23.1	32.4	7.8	20.1	8.3	0.57	15.4
2.24	735.066	11	6.2	274	23.1	32.7	7.8	20.1	8.3	0.57	15.4
3.36	735.048	10.9	6.3	273	23.1	32.6	7.8	20.1	8.3	0.57	15.4
8.96	735.027	10.9	6.3	271	23.2	32.3	7.9	20.1	8.3	0.57	15.4
13.44	735.023	10.8	6.4	270	23.3	32.2	7.8	20.1	8.3	0.57	15.4

Abbreviations: OFV, objective functional value; CL/F, apparent oral clearance; V/F, apparent volume of distribution; K_a , absorption rate constant; $\omega_{CL/F}$, inter-individual variability of CL/F; σ_1 , residual variability, proportional error; σ_2 , residual variability, additive error; RSE, relative standard error.

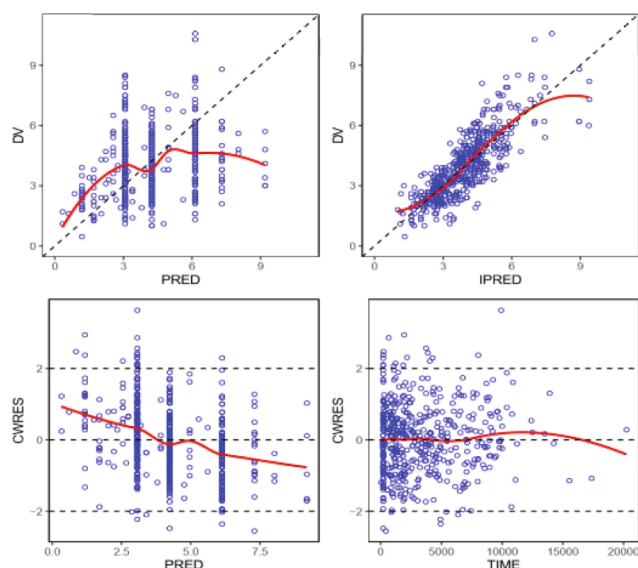
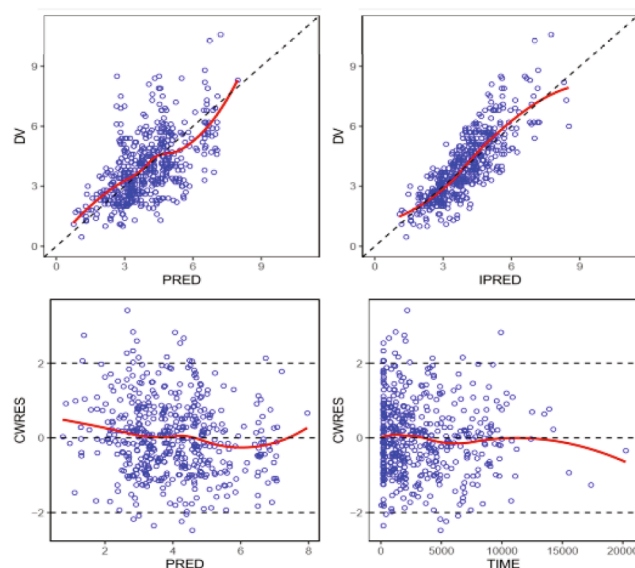
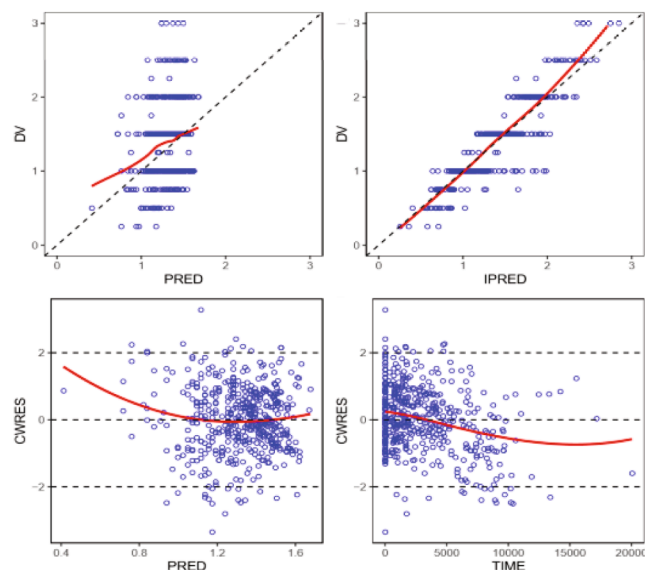
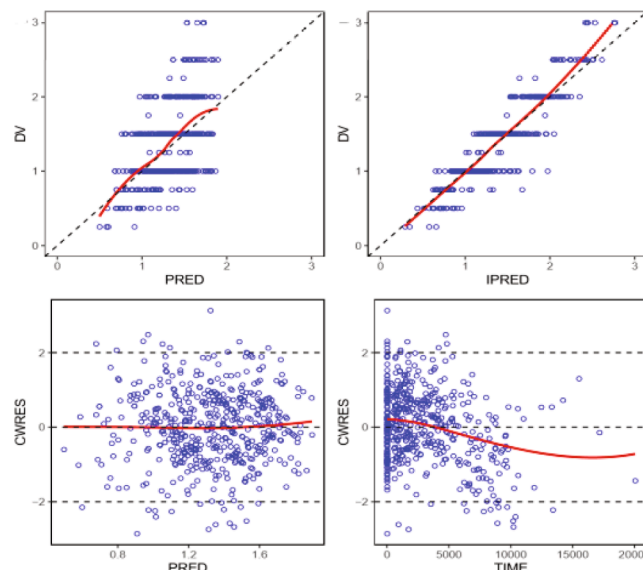
A. one-compartment model**A1-base model of linear model****A2-final model of linear model****B. Michaelis-Menten model****B1-base model of non-linear model****B2-final model of non-linear model**

Fig. 2. Goodness-of-fit plots of the (A) one-compartment model and (B) Michaelis-Menten model. DV, observed value; PRED, population predicted value; IPRED, individual predicted value; CWRES, conditional weighted residuals.

base model were 10.9 L/h and 272 L, respectively.

In the forward inclusion step, CYP3A5 genotype, DD, WT, daily dose of diltiazem and AST were retained in the model, and the AST was excluded in the backward elimination step. The final model was as follows:

$$K_a \text{ (h}^{-1}\text{)} = 4.48$$

$$CL/F \text{ (L/h)} = 14.2 \times \left(\frac{DD}{1.5}\right)^{0.201} \times \left(\frac{WT}{20}\right)^{0.173} \times \exp\left(\left(\frac{DILT}{30}\right) \times (-0.0417)\right) \times (1 + \theta_{CYP3A5})$$

$$V/F \text{ (L)} = 172$$

where $\theta = 0$ for CYP3A5*1*1, $\theta = -0.198$ for CYP3A5*1*3, and $\theta = -0.403$ for CYP3A5*3*3.

It is noteworthy that the daily dose of TAC retained in the final model. The nonlinear correlation between the daily dose of TAC and CL/F showed that the higher the daily dose, the higher the CL/F.

3.2.2. Nonlinear model (MM model)

The mixed residual model was employed, and the population typical values of V_{max} and K_m in the base MM model were 1.95 mg/day and 1.72 ng/ml, respectively. The effects of covariates on V_{max} and K_m were investigated. After stepwise screening, WT, CYP3A5*3 genotype, and co-therapy with diltiazem were retained as important covariates for K_m . The final MM model was as follows:

$$V_{max} \text{ (mg/day)} = 2.15$$

$$K_m \text{ (ng/ml)} = 0.845 \times \left(\frac{WT}{20}\right)^{-0.521} \times \exp\left(\left(\frac{DILT}{30}\right) \times 0.184\right) \times (1 + \theta_{CYP3A5})$$

where $\theta = 0$ for CYP3A5*1*1, $\theta = 1.63$ for CYP3A5*1*3, and $\theta = 4.0$ for CYP3A5*3*3.

Table 4

Parameter estimates of the base model, final model and bootstrap under two modeling strategies.

Parameter	Base mode		Final mode		Bootstrap (1000 times)		Bias (%)
	Estimate	RSE (%)	Estimate	RSE (%)	Median	2.5%-97.5%	
Linear one-compartment model							
OFV	735.04		620.55				
CL/F(L/h)	10.9	6.3	14.2	11.1	14.2	13.1–15.3	0
V/F(L)	272	23.1	172	25.4	173	147–204	0.58
Ka(h ⁻¹)	4.48	/	4.48	/	4.48	/	/
CYP3A5*1*1	/	/	-0.198	21.3	-0.196	(-0.228) - (-0.167)	-1.01
CYP3A5*3*3	/	/	-0.403	9.6	-0.403	(-0.431) - (-0.376)	0
WT	/	/	0.173	27.4	0.175	0.138–0.207	1.56
DD	/	/	0.201	21.1	0.200	0.170–0.226	-0.50
DILT	/	/	-0.0417	41.2	-0.0422	(-0.0535) - (-0.0301)	1.20
ωCL/F (%)	32.4	7.8	16.7	15.4	16.2	14.6–18.0	-3.0
σ ₁ (%)	20.1	8.3	21.5	6.5	21.5	20.5–22.4	0
σ ₂	0.57	15.4	0.473	19.5	0.46	0.39–0.53	-2.75
Nonlinear Michaelis–Menten (MM) model							
OFV	-621.46		-690.30				
V _{max} (mg/day)	1.95	6.4	2.15	6.8	2.15	2.06–2.27	0
K _m (ng/ml)	1.72	18.5	0.845	26.9	0.850	0.68–1.02	0.59
CYP3A5*1*1	/	/	1.63	21.7	1.65	1.41–1.93	1.23
CYP3A5*3*3	/	/	4	22.0	3.97	3.36–4.69	-0.75
WT	/	/	-0.521	42.4	-0.526	(-0.670) - (-0.354)	0.96
DILT	/	/	0.184	38.4	0.178	0.133–0.227	-3.26
ωV _{max} (%)	27.5	8.8	24.1	11.1	23.7	21.9–25.5	-1.66
ωK _m (%)	58.4	16.1	30.3	35.7	29.0	19.9–37.3	-4.29
σ ₁ (%)	15.5	12.9	14.2	17.6	13.9	12.0–15.5	-2.11
σ ₂	0.109	33.9	0.127	28.5	0.128	0.094–0.152	0.79

Abbreviations: OFV, objective functional value; CL/F, apparent oral clearance; V/F, apparent volume of distribution; Ka, absorption rate constant; WT, weight; DD, daily dose of tacrolimus; DILT, daily dose of diltiazem; ωCL/F, inter-individual variability of CL/F; σ₁, residual variability, proportional error; σ₂, residual variability, additive error; V_{max}, maximum daily dose at the steady-state concentration; K_m, steady-state trough concentration at the half-maximum daily dose; ωV_{max}, inter-individual variability of V_{max}; ωK_m, inter-individual variability of K_m; RSE, relative standard error; Bias (%) = (Bootstrap Median-NONMEM Estimate) / Estimate × 100.

3.3. Model evaluation

The GOF for the linear and nonlinear models are shown in Fig. 2. Compared with the base model, the final model showed no obvious bias or significant trends within these scatter plots, which indicated the good fit of the final model.

The parameter estimates of the two models and bootstrap evaluation are presented in Table 4. The median values estimated by bootstrap were consistent with the respective values of the final model, with < 5% bias, demonstrating that the parameter estimates were accurate and the model was steady. The success rates determined using the bootstrap method for the nonlinear MM model and the linear model were 98.5% and 95.7%, respectively.

The predictive ability of the two models was also evaluated by NDPE analysis (Fig. 3). Quantile-quantile plots and histograms confirmed the normality of NPDE, and statistics showed that the prediction discrepancy followed a normal distribution, indicating the good predictability of the final model. The pcVPC for the two models is shown in Fig. 4. Most of the observed concentrations were included in the 95% prediction intervals from the simulation data, indicating a good predictive performance of the final model.

In addition, the IF₂₀ (82.4%) and IF₃₀ (91.7%) of the MM model reached > 80%, implying its superiority compared with the one-compartment model (for which IF₂₀ and IF₃₀ were 61.3% and 80.4%, respectively).

3.4. Dosing regimens

Monte Carlo simulations were carried out based on the nonlinear MM model due to the better predictive performance than that of the linear model. The required dose was calculated using the V_{max} and K_m values according to the desired target trough concentration of 5–10 ng/ml. CYP3A5 genotype, WT and daily dose of TAC were considered in the

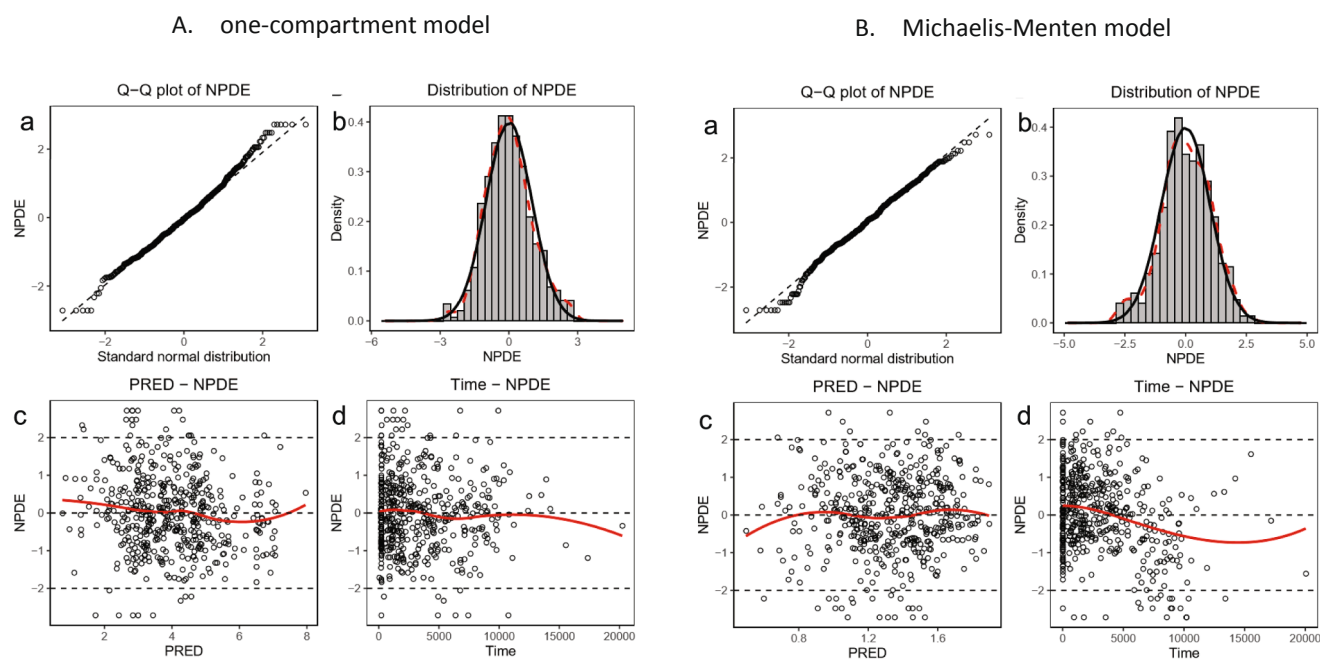


Fig. 3. Normalized prediction distribution errors (NPDE) of the (A) one-compartment model and (B) Michaelis-Menten model. a: Q-Q plot of NPDE; b: Distribution of NPDE; c: NPDE vs. population predicted concentration (PRED); d: NPDE vs. time after the last dose.

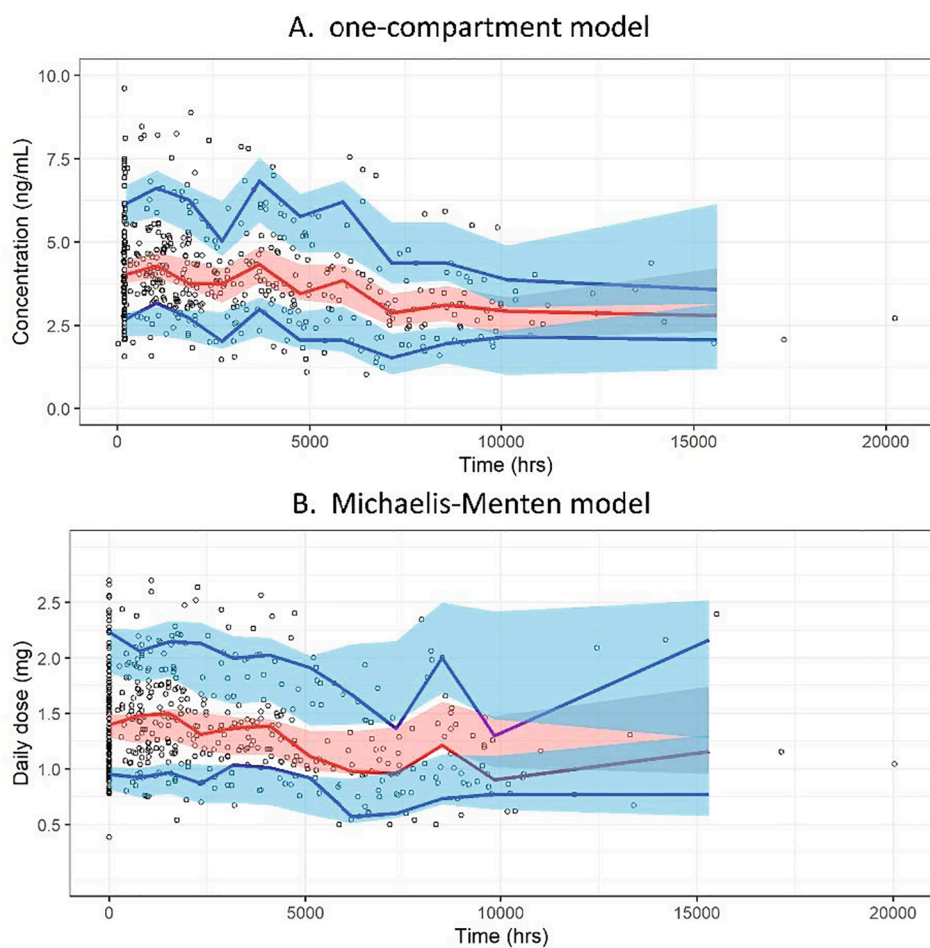


Fig. 4. Prediction-corrected visual predictive check (pcVPC) of the (A) one-compartment model and (B) Michaelis-Menten model. The middle red dashed line represents the median of the prediction-corrected concentrations. The lower and upper blue dashed lines are the 2.5th and 97.5th percentiles of the prediction-corrected concentrations, respectively.

simulation. Two thousand simulations were performed for each clinical setting. The results of the simulation are listed in Fig. 5. To achieve the target concentration, *CYP3A5**1*1 carriers required a lower maintenance dose than *CYP3A5**1*3 and *3*3 carriers. With an increase in WT, a higher WT-normalized TAC dose was required, and the use of diltiazem was able to reduce the dose requirement for TAC.

4. Discussion

With PPK, pharmacokinetic information can be obtained from sparse data collected from a large number of patients. Restricted by ethics and other factors, excessive blood sample collection is difficult in children [36,37]. Therefore, we established a PPK model of TAC in paediatric RNS by collecting trough concentrations obtained from TDM. This retrospective PPK modelling study of TAC in 108 children with RNS represents the largest study of its type in this patient category to date, and it's more extensive representation of the studied population.

In most previously published studies, the linear compartmental model was chosen, which is consistent with the traditional viewpoint that the PK of TAC is linear. We established a one-compartment linear model, K_a was set as 4.48 h^{-1} according to the literature for the reason of missing concentrations in the absorption and distribution phase. Interestingly, we also found that the daily dose of TAC was retained in the final model, and its inclusion improved the predictive performance of the final model significantly. Zhao et al. conducted an external evaluation of the published PPK models of TAC in adult renal transplant recipients, the published models were unsatisfactory in prediction- and simulation-based diagnostics [38]. They also found that the predictability of the three linear PK models was significantly improved with the incorporation of TAC dose, the MM model showed superior performance, which further supporting the potential nonlinear dynamics of TAC.

Therefore, we considered that there may be a nonlinear process in TAC steady-state PK, and established a nonlinear MM model in children with RNS. Only *CYP3A5* genotype, WT and daily dose of diltiazem were retained in the final MM model. We found that the MM model was better fit for the data compared with the one-compartment model for it demonstrated a better individual prediction. This finding is consistent with Huang's previous population pharmacokinetics analysis of TAC in children with NS [19]. The incorporation of non-linear pharmacokinetics in the establishment of TAC PPK model in children with RNS might be a promising approach to improving model predictability.

In the final MM model, the typical value of $V_{\max}$ was 2.15 mg/day, which indicated that the daily dose of TAC in excess of 2.15 mg may lead

to an unexpected increase in concentration. The typical value of K_m was 0.845 ng/ml, which indicated that when C_{ss} exceeded 0.845 ng/ml, the nonlinearity of TAC would affect dose rate adjustments. The value of $V_{\max}$ was the same as reported by Huang et al. [19], while the value of K_m was significantly lower than in their report (K_m was 1.98 ng/ml). This difference might be due to the lower trough concentrations of TAC in our study ($3.73 \pm 1.6\text{ ng/ml}$, ranging from 2 ng/ml to 10.6 ng/ml versus $5.0 \pm 2.9\text{ ng/ml}$, ranging from 2 ng/ml to 29.2 ng/ml). These lower concentrations of TAC have been proven effective in maintaining remission, as shown in our previous report [32]. In addition, the average age of the children in our study was lower than in Huang et al. (5.5 years vs. 7.8 years), and various developmental changes in children, including relative liver WT, immature liver enzymes and hepatic blood flow, etc., may influence the nonlinear metabolic processes of TAC, especially in children younger than 3 years of age.

At present, the initial dose of TAC for the treatment of NS in children is mostly WT-dependent, and Kidney disease: Improving global outcomes (KDIGO) clinical practice guideline is recommended at 0.1 mg/kg/day divided into two doses [39]. The effect of WT on TAC pharmacokinetics is more pronounced in paediatric patients than in adults. We found that WT was positively correlated with CL/F in the linear model and negatively correlated with K_m in the nonlinear model. With an increase in WT, a higher WT-normalized TAC dose was required. This phenomenon is consistent with other population pharmacokinetic studies of children with NS [15,19]. In the population pharmacokinetics study of adult NS patients, WT was not associated with TAC clearance [40]. The explanation for the effect of WT on TAC metabolism is not clear, though it may be related to children's growth and development processes. Hepatic blood flow and relative quality may influence liver clearance, both of which increase in children when normalized by WT [41]. In addition, changes in liver metabolic enzyme activity may also affect TAC metabolism [42].

Meanwhile, *CYP3A5* genotype, rather than WT, significantly affected the PK of TAC in children with RNS both in the linear and nonlinear models, suggesting that much of the PK variability in TAC was not WT-dependent. Tacrolimus is metabolized primarily by cytochrome P450 (CYP) 3A enzymes, and there have been numerous reports suggesting that polymorphisms in *CYP3A5* affect TAC clearance [43–46]. The Clinical Pharmacogenetics Implementation Consortium (CPIC) guidelines for TAC use in transplant patients recommend that, compared to *CYP3A5**3*3 carriers, the initial dose of TAC in patients who are *CYP3A5**1*1 or *1*3 carriers should be increased by 1.5–2 times [26]. However, in Caucasians, *CYP3A5**1*1 only accounts for 1% of the population, while it accounts for approximately 7.7% to 11.1% of the

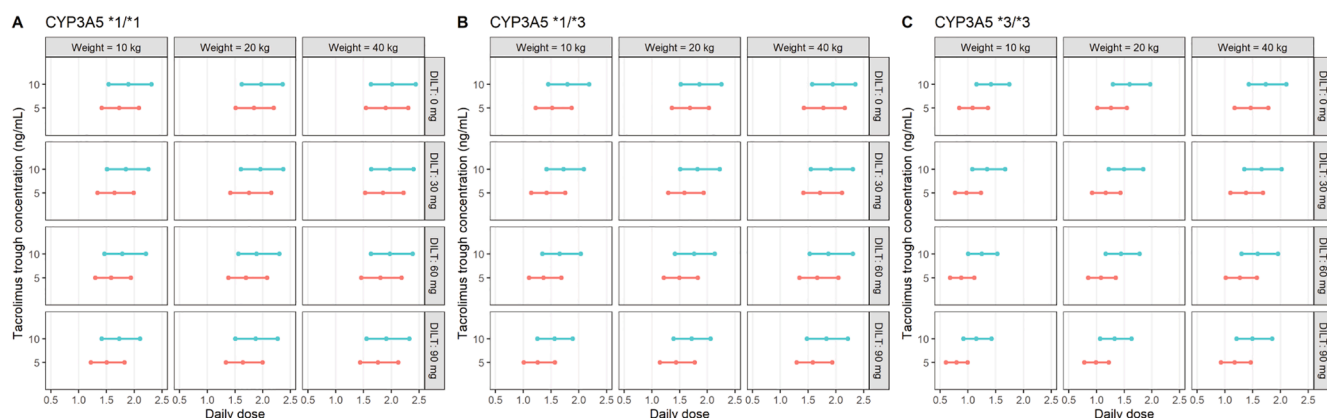


Fig. 5. Simulations of the predicted dose of tacrolimus at the target trough concentration. DILT: diltiazem. A, B and C represent the genotypes of *CYP3A5* were *CYP3A5**1*1, *CYP3A5**1*3 and *CYP3A5**3*3, respectively. The vertical axis represents the target concentration of TAC were 5 ng/ml and 10 ng/ml. The horizontal axis was the daily dose of tacrolimus required. Each column represents a body weight of 10 kg, 20 kg and 40 kg. Each row represents a different dose of diltiazem in combination. The red and blue lines represent the daily dose of TAC required at the target concentration of 5 ng/ml and 10 ng/ml, respectively. The dots on each line represent, from left to right, the 25, 50, and 75 percentiles.

Chinese population and approximately 37% to 45% of the Black population [32,47]. To the best of our knowledge, this is the first PPK model of TAC in children with RNS that distinguishes between the effects of wild type (*1*1), heterozygous type (*1*3) and homozygous type (*3*3). The numbers of children in this study with *CYP3A5**1*1, *1*3 and *3*3 genotypes were 12, 50 and 46, respectively. The CL/F of *CYP3A5**1*1 carriers was 1.25-fold higher than that of *CYP3A5**1*3 carriers and 1.68-fold higher than that of *CYP3A5**3*3 carriers. The K_m of *CYP3A5**1*3 carriers was 2.63-fold higher than that of *CYP3A5**1*1 carriers, and the K_m of *CYP3A5**3*3 carriers was 1.9-fold higher than that of *CYP3A5**1*3 carriers. It is notable that a higher dose of TAC was needed in *CYP3A5**1*1 carriers than in the other carriers. A *CYP3A5* genotype-based dosing regimen might be a useful approach to better predict individual TAC doses in children with RNS and to achieve the target therapeutic concentration faster, especially in Chinese children.

Diltiazem has been reported to be an effective antihypertensive agent for children in doses ranging from 3 to 6 mg/kg/day and it is well-tolerated [48]. Moreover, diltiazem is also frequently used as a TAC-sparing agent in the treatment of children with NS in our hospital, which can increase the concentration of TAC by inhibiting the *CYP3A4* enzyme, thereby saving on medical costs, controlling blood pressure and improving renal function [27,32,49,50]. M Gokce et al. found that diltiazem can effectively alleviate renal ischemia reperfusion injury in rats [51]. Wenjing Wang et al. pointed out that the lower total dose of TAC may be partly responsible for the lower incidence of nephrotoxicity, while the combination of diltiazem can reduce the total dose of TAC [52]. Many of the children included in this study were routinely administered diltiazem with their first dose of TAC or when their TAC blood concentrations were low. Diltiazem was given at doses ranging from 1 to 4 mg/kg/day, which varies widely but was less than the reported doses for the treatment of hypertension. The narrow therapeutic window and poor dose-concentration correlation for TAC make it difficult to select the dose when diltiazem is used in combination to increase the plasma concentration of TAC clinically. Actually, in children who co-administration with diltiazem, nearly 46% of them had adjusted their dose of diltiazem because of the unsatisfactory concentration of TAC. In the final MM model, K_m increased by 20%, 44%, and 74% with a daily dose of 30, 60, and 90 mg of diltiazem, respectively, compared to those did not receive diltiazem.

From Fig. 5, we found that the combination of diltiazem required a lower WT-normalized TAC dose. It is worth noting that, this phenomenon was more evident in *CYP3A5**3*3 carriers than in other carriers. *CYP3A5* is the main metabolic enzyme of TAC, the affinity of *CYP3A5* was similar to that of *CYP3A4*, while the catalytic efficiency (V_{max}/K_m) was 64% higher than that of *CYP3A4* [53,54]. However, the relative contribution of *CYP3A5* strongly depends on the concomitant amount of *CYP3A4*. When *CYP3A5* loses its expression and function (*CYP3A5* non-expressers, *CYP3A5**3*3), *CYP3A4* will take over the metabolic processing of TAC [25]. Therefore, the reason for this might be that when *CYP3A5* is disabled, the combined use of diltiazem inhibits the activity of *CYP3A4*, resulting in a significant reduction in TAC metabolism. Another noteworthy point regarding the use of diltiazem is that because TAC is prescribed at 0.5 mg per capsule, it is difficult to give an accurate dose when the required daily TAC dose is not an integer multiple of 0.5, while the daily dose of diltiazem can be controlled. For example, as shown in Fig. 5, for a child weighing 20 kg with the *CYP3A5**1*3 genotype, the daily dose of TAC is approximately 1.7 mg for a target concentration of 5 ng/ml. However, practically speaking, it is difficult to administer this dose. If 2 mg is administered, the TAC concentration in children may be too high, leading to adverse reactions such as nephrotoxicity. At this point, 1.5 mg of TAC combined with 60 mg of diltiazem would be clinically recommended for this child. It can effectively avoid excessive drug concentration and can protect the kidneys at the same time.

There were some limitations in our study. First, it was a single-centre retrospective study, and some potentially important indicators, such as

haematocrit, were not included as covariates for some children lack data and were difficult to fill artificially. Second, berberine, azole antifungal agents, macrolide antibiotics, Wuzhi capsules and simvastatin had also been reported to affect TAC metabolism [33,55–59]. However, very few children in our hospital use these drugs in combination, so only diltiazem was considered for covariate inclusion. Finally, only CL/F and its covariate could be reliably assessed because of lacking concentrations in the absorbed and distributed phases. For the same reason, the function of daily dose might mainly reflect the nonlinearity of the clearance process, and it is difficult to elucidate the specific nonlinear PK processes. Further investigation will be needed.

5. Conclusions

In conclusion, this study presents one-compartment linear and nonlinear PPK models of TAC that make use of a large body of data collected from children with RNS. The nonlinear model better described the pharmacokinetic behaviour of TAC than the linear model. *CYP3A5* genotype, WT and daily dose of diltiazem were identified as significant covariates in the final nonlinear MM model. The target concentration of TAC can be better achieved by adjusting the dosage of diltiazem appropriately. The exact mechanism of the nonlinearity of TAC is still unknown, and it is impossible to elucidate only with the trough concentration. Further prospective investigations are needed.

CRedit authorship contribution statement

Ling Li: Formal analysis, Methodology, Software, Writing – original draft. **Min Zhu:** Data curation, Formal analysis, Software, Validation. **De-Yi Li:** Data curation, Methodology, Software. **Hong-Li Guo:** Investigation, Methodology. **Ya-Hui Hu:** Investigation, Data curation. **Ze-Yue Xu:** Investigation, Data curation. **Xia Jing:** Funding acquisition. **Feng Chen:** Conceptualization, Funding acquisition, Writing – review & editing. **Fei Zhao:** Conceptualization, Resources, Methodology. **Yun-Man Li:** Supervision, Writing – review & editing. **Jing Xu:** Funding acquisition, Supervision, Writing – review & editing. **Zheng Jiao:** Validation, Supervision, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Medical Science Project of Jiangsu Commission of Health [Z2020040]; the Specially Appointed Medical Expert Project of Jiangsu Commission of Health [2019]; the Medical Science and Technique Foundation of Nanjing Health Commission [YKK18147]; and the Hospital Pharmacy Foundation of Jiangsu Pharmaceutical Association [A201810].

References

- [1] M.L. Downie, C. Gallibois, R.S. Parekh, D.G. Noone, Nephrotic syndrome in infants and children: pathophysiology and management, *Paediatr. Int. Child Health* 37 (4) (2017) 248–258.
- [2] R.M. Lombel, D.S. Gipson, E.M. Hodson, O. Kidney Disease: Improving Global, Treatment of steroid-sensitive nephrotic syndrome: new guidelines from KDIGO, *Pediatric Nephrol.* 28(3) (2013) 415–426.
- [3] H.X. Chen, Q. Cheng, F. Li, Q.N. He, Y. Cao, Z.W. Yi, X.C. Wu, Efficacy and safety of tacrolimus and low-dose prednisone in Chinese children with steroid-resistant nephrotic syndrome, *WJP, World journal of pediatrics*, 2019.
- [4] E.M. Yang, S.T. Lee, H.J. Choi, H.Y. Cho, J.H. Lee, H.G. Kang, Y.S. Park, H. I. Cheong, I.S. Ha, Tacrolimus for children with refractory nephrotic syndrome: a one-year prospective, multicenter, and open-label study of Tacrobell(R), a generic formula, *World J. Pediatr.* 12 (1) (2016) 60–65.

- [5] M. Colucci, G. Corpetti, F. Emma, M. Vivarelli, Immunology of idiopathic nephrotic syndrome, *Pediatr. Nephrol.* 33 (4) (2018) 573–584.
- [6] M. Brunet, T. van Gelder, A. Asberg, V. Haulfroid, D.A. Hesselink, L. Langman, F. Lemaître, P. Marquet, C. Seger, M. Shipkova, A. Vinks, P. Wallemacq, E. Wieland, J.B. Woillard, M.J. Barten, K. Budde, H. Colom, M.T. Dieterlen, L. Elens, K.L. Johnson-Davis, P.K. Kunicki, I. MacPhee, S. Masuda, B.S. Mathew, O. Millan, T. Mizuno, D.A.R. Moes, C. Monchaud, O. Noceti, T. Pawinski, N. Picard, R. van Schaik, C. Sommerer, N.T. Vethe, B. de Winter, U. Christians, S. Bergan, Therapeutic drug monitoring of tacrolimus-personalized therapy: second consensus report, *Ther. Drug Monit.* 41 (3) (2019) 261–307.
- [7] J.M. Barbarino, C.E. Staats, R. Venkataramanan, T.E. Klein, R.B. Altman, PharmGKB summary: cyclosporine and tacrolimus pathways, *Pharmacogenet. Genomics* 23 (10) (2013) 563–585.
- [8] C. Faul, M. Donnelly, S. Merscher-Gomez, Y.H. Chang, S. Franz, J. Delfgaauw, J. M. Chang, H.Y. Choi, K.N. Campbell, K. Kim, J. Reiser, P. Mundel, The actin cytoskeleton of kidney podocytes is a direct target of the antiproteinuric effect of cyclosporine A, *Nat. Med.* 14 (9) (2008) 931–938.
- [9] L. Fan, Q. Liu, Y. Liao, Z. Li, Y. Ji, Z. Yang, J. Chen, J. Fu, J. Zhang, Y. Kong, P. Fu, T. Lou, Z. Liu, X. Yu, W. Chen, Tacrolimus is an alternative therapy option for the treatment of adult steroid-resistant nephrotic syndrome: a prospective, multicenter clinical trial, *Int. Urol. Nephrol.* 45 (2) (2013) 459–468.
- [10] Y. Wen, L. Liu, P. Zhou, H. Li, Z. Wang, Y. Zhang, J. Liang, Tacrolimus restores podocyte injury and stabilizes the expression of Cabin1 in 5/6 nephrectomized rats, *Ren. Fail.* 38 (4) (2016) 564–570.
- [11] T.H. Nguyen, M.S. Mouksassi, N. Holford, N. Al-Huniti, I. Freedman, A.C. Hooker, J. John, M.O. Karlsson, D.R. Mould, J.J. Perez Ruiz, E.L. Plan, R. Savic, J.G. van Hasselt, B. Weber, C. Zhou, E. Comets, F. Mentre, C. Model Evaluation Group of the International Society of Pharmacometrics Best Practice, Model Evaluation of Continuous Data Pharmacometric Models: Metrics and Graphics, *CPT Pharmacometrics Syst. Pharmacol.* 6(2) (2017) 87–109.
- [12] D.L. Goodall, M. Willcombe, A.G. McLean, D. Taube, High inpatient variability of tacrolimus levels and outpatient clinic nonattendance are associated with inferior outcomes in renal transplant patients, *Transpl. Direct* 3 (8) (2017), e192.
- [13] J.E. Ericson, K.O. Zimmerman, D. Gonzalez, C. Melloni, J.T. Guptill, K.D. Hill, H. Wu, M. Cohen-Wolkowicz, A Systematic Literature Review Approach to Estimate the Therapeutic Index of Selected Immunosuppressant Drugs After Renal Transplantation, *Ther. Drug Monit.* 39 (1) (2017) 13–20.
- [14] J.B. Woillard, J. Debord, C. Monchaud, F. Saint-Marcoux, P. Marquet, Population Pharmacokinetics and Bayesian Estimators for Refined Dose Adjustment of a New Tacrolimus Formulation in Kidney and Liver Transplant Patients, *Clin. Pharmacokinet.* 56 (12) (2017) 1491–1498.
- [15] G.X. Hao, X. Huang, D.F. Zhang, Y. Zheng, H.Y. Shi, Y. Li, E. Jacqz-Aigrain, W. Zhao, Population pharmacokinetics of tacrolimus in children with nephrotic syndrome, *Br. J. Clin. Pharmacol.* 84 (8) (2018) 1748–1756.
- [16] D. Wang, J. Lu, Q. Li, Z. Li, Population pharmacokinetics of tacrolimus in pediatric refractory nephrotic syndrome and a summary of other pediatric disease models, *Exp. Ther. Med.* 17 (5) (2019) 4023–4031.
- [17] X. Wang, Y. Han, C. Chen, L. Ma, H. Xiao, Y. Zhou, Y. Cui, F. Wang, B. Su, Y. Yao, J. Ding, Population pharmacokinetics and dosage optimization of tacrolimus in pediatric patients with nephrotic syndrome, *Int. J. Clin. Pharmacol. Ther.* 57 (3) (2019) 125–134.
- [18] X. Chen, D.D. Wang, H. Xu, Z.P. Li, Optimization of initial dosing scheme of tacrolimus in pediatric refractory nephrotic syndrome patients based on CYP3A5 genotype and coadministration with wuzhi-capsule, *Xenobiotica* 50 (5) (2020) 606–613.
- [19] L. Huang, Y. Liu, Z. Jiao, J. Wang, L. Fang, J. Mao, Population pharmacokinetic study of tacrolimus in pediatric patients with primary nephrotic syndrome: a comparison of linear and nonlinear Michaelis-Menten pharmacokinetic model, *Eur. J. Pharm. Sci.* 143 (2020), 105199.
- [20] A. Jahan, R. Prabha, S. Chaturvedi, B. Mathew, D. Fleming, I. Agarwal, Clinical efficacy and pharmacokinetics of tacrolimus in children with steroid-resistant nephrotic syndrome, *Pediatr. Nephrol.* 30 (11) (2015) 1961–1967.
- [21] R.M. Lombel, E.M. Hodson, D.S. Gipson, O. Kidney Disease: Improving Global, Treatment of steroid-resistant nephrotic syndrome in children: new guidelines from KDIGO, *Pediatr. Nephrol.* 28(3) (2013) 409–414.
- [22] G.X. Hao, L.L. Song, D.F. Zhang, L.Q. Su, E. Jacqz-Aigrain, W. Zhao, Off-label use of tacrolimus in children with glomerular disease: Effectiveness, safety and pharmacokinetics, *Br. J. Clin. Pharmacol.* 86 (2) (2020) 274–284.
- [23] M. Yu, M. Liu, W. Zhang, Y. Ming, Pharmacokinetics, Pharmacodynamics and Pharmacogenetics of Tacrolimus in Kidney Transplantation, *Curr. Drug Metab.* 19 (6) (2018) 513–522.
- [24] R. Mac Guad, N.L. Zaharan, Z. Chik, Z. Mohamed, N.K. Peng, W.A. Adnan, Effects of CYP3A5 Genetic Polymorphism on the Pharmacokinetics of Tacrolimus in Renal Transplant Recipients, *Transpl. Proc.* 48 (1) (2016) 81–87.
- [25] J.Y. Sun, Z.J. Xu, F. Sun, H.L. Guo, X.S. Ding, F. Chen, J. Xu, Individualized Tacrolimus Therapy for Pediatric Nephrotic Syndrome: Considerations for Ontogeny and Pharmacogenetics of CYP3A, *Curr. Pharm. Des.* 24 (24) (2018) 2765–2773.
- [26] K.A. Birdwell, B. Decker, J.M. Barbarino, J.F. Peterson, C.M. Stein, W. Sadee, D. Wang, A.A. Vinks, Y. He, J.J. Swen, J.S. Leeder, R. van Schaik, K.E. Thummel, T. E. Klein, K.E. Caudle, I.A. MacPhee, Clinical Pharmacogenetics Implementation Consortium (CPIC) Guidelines for CYP3A5 Genotype and Tacrolimus Dosing, *Clin. Pharmacol. Ther.* 98 (1) (2015) 19–24.
- [27] C.L. Choong, H.S. Wong, F.Y. Lee, C.K. Lee, J.V. Kho, Y.X. Lai, T. Vikneswaran, Dose-Response Relationship Between Diltiazem and Tacrolimus and Its Safety in Renal Transplant Recipients, *Transpl. Proc.* 50 (8) (2018) 2515–2520.
- [28] M.F. Hebert, A.Y. Lam, Diltiazem increases tacrolimus concentrations, *Ann. Pharmacother.* 33 (6) (1999) 680–682.
- [29] J.L. Li, X.D. Wang, S.Y. Chen, L.S. Liu, Q. Fu, X. Chen, L.C. Teng, C.X. Wang, M. Huang, Effects of diltiazem on pharmacokinetics of tacrolimus in relation to CYP3A5 genotype status in renal recipients: from retrospective to prospective, *Pharmacogenomics J.* 11 (4) (2010) 300–306.
- [30] J.Y. Sun, Y.H. Hu, H.L. Guo, X. Jing, Z.J. Xu, F. Sun, H.L. Guo, X.S. Ding, F. Chen, J. Xu, Diltiazem used as a tacrolimus-sparing agent for treatment of pediatric patients with refractory nephrotic syndrome: a case report and retrospective analysis, *Eur. J. Clin. Pharmacol.* 75 (4) (2019) 591–593.
- [31] D.G. Noone, K. Iijima, R. Parekh, Idiopathic nephrotic syndrome in children, *Lancet* 392 (10141) (2018) 61–74.
- [32] H.L. Guo, J. Xu, J.Y. Sun, L. Li, H.L. Guo, X. Jing, Z.Y. Xu, Y.H. Hu, Z.J. Xu, F. Sun, X.S. Ding, F. Chen, F. Zhao, Tacrolimus treatment in childhood refractory nephrotic syndrome: A retrospective study on efficacy, therapeutic drug monitoring, and contributing factors to variable blood tacrolimus levels, *Int. Immunopharmacol.* 81 (2020), 106290.
- [33] D.D. Wang, J.M. Lu, Q. Li, Z.P. Li, Population pharmacokinetics of tacrolimus in paediatric systemic lupus erythematosus based on real-world study, *J. Clin. Pharm. Ther.* 43 (4) (2018) 476–483.
- [34] D.D. Wang, X. Chen, Z.P. Li, Wuzhi capsule and haemoglobin influence tacrolimus elimination in paediatric kidney transplantation patients in a population pharmacokinetics analysis: a retrospective study, *J. Clin. Pharm. Ther.* 44 (4) (2019) 611–617.
- [35] J.W. Yang, S.S. Liao, L.Q. Zhu, Y. Zhao, Y. Zhang, X.Y. Sun, W. Rao, W. Qu, W.Z. Li, L.Y. Sun, Population pharmacokinetic analysis of tacrolimus early after Chinese paediatric liver transplantation, *Int. J. Clin. Pharmacol. Ther.* 53 (1) (2015) 75–83.
- [36] R.E. Kauffman, G.L. Kearns, Pharmacokinetic studies in paediatric patients, *Clin. Ethical Considerations Clin. Pharmacokinetics* 23 (1) (1992) 10–29.
- [37] F. Saint-Marcoux, J.B. Woillard, C. Jurado, P. Marquet, Lessons from routine dose adjustment of tacrolimus in renal transplant patients based on global exposure, *Ther. Drug Monit.* 35 (3) (2013) 322–327.
- [38] C.Y. Zhao, Z. Jiao, J.J. Mao, X.Y. Qiu, External evaluation of published population pharmacokinetic models of tacrolimus in adult renal transplant recipients, *Br. J. Clin. Pharmacol.* 81 (5) (2016) 891–907.
- [39] S. Samuel, M. Bitzan, M. Zappitelli, A. Dart, C. Mammen, M. Pinski, A.V. Cybulsky, M. Walsh, G. Knoll, M. Hladunewich, J. Bargman, H. Reich, A. Humar, N. Muirhead, Canadian Society of Nephrology Commentary on the 2012 KDIGO clinical practice guideline for glomerulonephritis: management of nephrotic syndrome in children, *Am. J. Kidney Dis.* 63 (3) (2014) 354–362.
- [40] T. Lu, X. Zhu, S. Xu, M. Zhao, X. Huang, Z. Wang, L. Zhao, Dosage Optimization Based on Population Pharmacokinetic Analysis of Tacrolimus in Chinese Patients with Nephrotic Syndrome, *Pharm. Res.* 36 (3) (2019) 45.
- [41] D.J. Murry, W.R. Crom, W.E. Reddick, R. Bhargava, W.E. Evans, Liver volume as a determinant of drug clearance in children and adolescents, *Drug. Metab. Dispos.* 23 (10) (1995) 1110–1116.
- [42] S.N. de Wildt, G.L. Kearns, J.S. Leeder, J.N. van den Anker, Cytochrome P450 3A: ontogeny and drug disposition, *Clin. Pharmacokinet* 37 (6) (1999) 485–505.
- [43] T. Nioka, A. Komatsuda, S. Kato, M. Togashi, S. Okuyama, A. Omokawa, M. Nara, H. Wakui, N. Takahashi, M. Miura, Effects of CYP3A5 polymorphism and the tacrolimus 12 h concentration on tacrolimus-induced acute renal dysfunction in patients with lupus nephritis, *Xenobiotica Fate Foreign Compounds Biol. Syst.* 45 (12) (2015) 1147–1153.
- [44] H. Kato, M. Usui, Y. Muraki, A. Tanemura, Y. Murata, N. Kuriyama, Y. Azumi, M. Kishiwada, S. Mizuno, H. Sakurai, M. Okuda, K. Nakatani, S. Isaji, Long-term influence of CYP3A5 gene polymorphism on pharmacokinetics of tacrolimus and patient outcome after living donor liver transplantation, *Transpl. Proc.* 48 (4) (2016) 1087–1094.
- [45] A.A. Prytula, K. Cransberg, A.H. Bouts, R.H. van Schaik, H. de Jong, S.N. de Wildt, R.A. Muthof, The effect of weight and CYP3A5 genotype on the population pharmacokinetics of tacrolimus in stable paediatric renal transplant recipients, *Clin. Pharmacokinet* 55 (9) (2016) 1129–1143.
- [46] C.O. Jacobo-Cabral, P. Garcia-Roca, E.M. Romero-Tejeda, H. Reyes, M. Medeiros, G. Castaneda-Hernandez, I.F. Troconiz, Population pharmacokinetic analysis of tacrolimus in Mexican paediatric renal transplant patients: role of CYP3A5 genotype and formulation, *Br. J. Clin. Pharmacol.* 80 (4) (2015) 630–641.
- [47] L. Chen, G.V.R. Prasad, CYP3A5 polymorphisms in renal transplant recipients: influence on tacrolimus treatment, *Pharmacogenomics Pers. Med.* 11 (2018) 23–33.
- [48] J.T. Flynn, D.A. Pasko, Calcium channel blockers: pharmacology and place in therapy of pediatric hypertension, *Pediatr. Nephrol.* 15 (3–4) (2000) 302–316.
- [49] J.L. Li, X.D. Wang, S.Y. Chen, L.S. Liu, Q. Fu, X. Chen, L.C. Teng, C.X. Wang, M. Huang, Effects of diltiazem on pharmacokinetics of tacrolimus in relation to CYP3A5 genotype status in renal recipients: from retrospective to prospective, *Pharmacogenomics J.* 11 (4) (2011) 300–306.
- [50] L.Y. Zhou, X.C. Zuo, K. Chen, J.L. Wang, Q.J. Chen, Y.N. Zhou, H. Yuan, Y. Ma, L. J. Zhu, Y.X. Peng, Y.Z. Ming, Significant impacts of CYP3A4*1G and CYP3A5*3 genetic polymorphisms on the pharmacokinetics of diltiazem and its main metabolites in Chinese adult kidney transplant patients, *J. Clin. Pharm. Ther.* 41 (3) (2016) 341–347.
- [51] M. Gokce, M.F. Yuzbasioglu, E. Bulbuloglu, H. Oksuz, S. Yormaz, O. Altinoren, M. Kutlucan, I. Coskuner, E. Silay, I.T. Kale, Cilostazol and diltiazem attenuate cyclosporine-induced nephrotoxicity in rats, *Transplant Proc.* 44 (6) (2012) 1738–1742.
- [52] W. Wang, Y. Xia, J. Mao, Y. Chen, D. Wang, H. Shen, H. Fu, L. Du, A. Liu, Treatment of tacrolimus or cyclosporine A in children with idiopathic nephrotic syndrome, *Pediatr. Nephrol.* 27 (11) (2012) 2073–2079.

- [53] Y. Dai, M.F. Hebert, N. Isoherranen, C.L. Davis, C. Marsh, D.D. Shen, K. E. Thummel, Effect of CYP3A5 polymorphism on tacrolimus metabolic clearance in vitro, *Drug Metab. Dispos.* 34 (5) (2006) 836–847.
- [54] L.K. Kamdem, F. Streit, U.M. Zanger, J. Brockmoller, M. Oellerich, V. W. Armstrong, L. Wojnowski, Contribution of CYP3A5 to the in vitro hepatic clearance of tacrolimus, *Clin. Chem.* 51 (8) (2005) 1374–1381.
- [55] Q. Hou, W. Han, X. Fu, Pharmacokinetic interaction between tacrolimus and berberine in a child with idiopathic nephrotic syndrome, *Eur. J. Clin. Pharmacol.* 69 (10) (2013) 1861–1862.
- [56] E. Khan, M. Killackey, D. Kumbala, H. LaGuardia, Y.J. Liu, H.Z. Qin, B. Alper, A. Paramesh, J. Buell, R. Zhang, Long-term outcome of ketoconazole and tacrolimus co-administration in kidney transplant patients, *World J. Nephrol.* 3 (3) (2014) 107–113.
- [57] X. Chen, D.D. Wang, H. Xu, Z.P. Li, Optimization of initial dosing scheme of tacrolimus in pediatric refractory nephrotic syndrome patients based on CYP3A5 genotype and coadministration with wuzhi-capsule, *Xenobiotica Fate Foreign Compounds Biol. Syst.* (2019) 1–8.
- [58] V.J. Lempers, L.C. Martial, M.F. Schreuder, N.M. Blijlevens, D.M. Burger, R. E. Aarnoutse, R.J. Bruggemann, Drug-interactions of azole antifungals with selected immunosuppressants in transplant patients: strategies for optimal management in clinical practice, *Curr. Opin. Pharmacol.* 24 (2015) 38–44.
- [59] I.W. Kim, Y.J. Moon, E. Ji, K.I. Kim, N. Han, S.J. Kim, W.G. Shin, J. Ha, J.H. Yoon, H.S. Lee, J.M. Oh, Clinical and genetic factors affecting tacrolimus trough levels and drug-related outcomes in Korean kidney transplant recipients, *Eur. J. Clin. Pharmacol.* 68 (5) (2012) 657–669.